

Ex 4:

```
import java.math.BigInteger;
import java.util.Scanner;
public class Rsa {
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);
        int p = 3;
        int q = 11;
        int n = p * q;
        int phi = (p - 1) * (q - 1);
        int e = 7;
        int d = 1;
        while ((d * e) % phi != 1) {
            d++;
        }
        System.out.println("Public Key = (" + e + ", " + n + ")");
        System.out.println("Private Key = (" + d + ", " + n + ")");
        System.out.print("Enter message (number): ");
        int m = sc.nextInt();

        BigInteger M = BigInteger.valueOf(m);
        BigInteger C = M.pow(e).mod(BigInteger.valueOf(n));
        System.out.println("Encrypted message = " + C);
        BigInteger D = C.pow(d).mod(BigInteger.valueOf(n));
        System.out.println("Decrypted message = " + D);
    }
}
```

Output:

```
Public Key = (7, 33)
Private Key = (3, 33)
Enter message (number): 25
Encrypted message = 31
Decrypted message = 25
```

Ex 5:

```
import java.util.Scanner;
import java.math.BigInteger;
public class DiffieHellman {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter prime number q: ");
        int q = sc.nextInt();
        System.out.print("Enter primitive root a: ");
        int a = sc.nextInt();
        System.out.print("Enter private key XA: ");
        int XA = sc.nextInt();
        System.out.print("Enter private key XB: ");
        int XB = sc.nextInt();
        BigInteger Q = BigInteger.valueOf(q);
        BigInteger A = BigInteger.valueOf(a);
        int YA = A.modPow(BigInteger.valueOf(XA), Q).intValue();
        int YB = A.modPow(BigInteger.valueOf(XB), Q).intValue();
        int KA = BigInteger.valueOf(YB)
            .modPow(BigInteger.valueOf(XA), Q)
            .intValue();
        int KB = BigInteger.valueOf(YA)
            .modPow(BigInteger.valueOf(XB), Q)
            .intValue();
        System.out.println("Public key YA = " + YA);
        System.out.println("Public key YB = " + YB);
        System.out.println("Secret key of A = " + KA);
        System.out.println("Secret key of B = " + KB);
        if (KA == KB)
            System.out.println("Key Exchange Successful");
        else
            System.out.println("Key Exchange Failed");
        sc.close();
    }
}
```

Output:

```
Enter prime number q: 7
Enter primitive root a: 3
Enter private key XA: 5
Enter private key XB: 2
Public key YA = 5
Public key YB = 2
Secret key of A = 4
Secret key of B = 4
Key Exchange Successful
```